Available online at www.sciencedirect.com

ScienceDirect

Procedia Computer Science 256 (2025) 1208–1215

Procedia Computer Science

www.elsevier.com/locate/procedia

CENTERIS – International Conference on ENTERprise Information Systems / ProjMAN – International Conference on Project MANagement / HCist – International Conference on Health and Social Care Information Systems and Technologies 2024

Mobile Applications and Cloud Computing in Healthcare

Mihnea-Vlad Dan, Mădălin-Dorin Pop*

Politehnica University of Timişoara, Piaţa Victoriei 2, Timişoara 300006, Romania

Abstract

With the new advancements in cloud computing and internet of things (IoT), mobile applications are transforming the healthcare sector offering solutions for both users and researchers. The massive amounts of data generated by these kinds of applications can be used by researchers to reduce medical errors, for disease prevention, real-time alerts and patient monitoring. The aim of this paper is to present a solution for a mobile application for fitness and dieting, which can help users achieve a healthier lifestyle, through an interactive approach. The technologies used to achieve the goal are Flutter SDK for the frontend, allowing for an interactive, cross-platform application and Firebase for the backend and database services, a cloud solution allowing for data gathering that could be used in analysis and diagnosis. The article's scientific contribution includes not only helping in the process of gathering pertinent data for healthcare analysis through the mobile applications, but also reviewing previous research done in the field.

Peer-review under responsibility of the scientific committee of the CENTERIS - International Conference on ENTERprise Information Systems / ProjMAN - International Conference on Project MANagement / HCist - International Conference on Health and Social Care Information Systems and Technologies

Keywords: mobile applications; cloud computing; healthcare.

1. Introduction

Nowadays cell phones play an important role in our lives, with over half of the website traffic coming from them. As well as being able to communicate with other people, at any time and from any distance they offer us a range of

* Corresponding author.
E-mail address: madalin.pop@upt.ro

Peer-review under responsibility of the scientific committee of the CENTERIS - International Conference on ENTERprise Information Systems / ProjMAN - International Conference on Project MANagement / HCist - International Conference on Health and Social Care Information Systems and Technologies
10.1016/j.procs.2025.02.230

applications that make our lives easier. However, there are also disadvantages to this way of life. With the development of technology, we have become lazier, spending more and more time in front of screens, becoming more sedentary which can be seriously damaging to our health [1].

The effects of unhealthy diets and sedentary lifestyles show up in the current epidemic of overweight and obesity among both children and among adults in OECD countries [2]. The rates for obesity and overweight people tend to increase with age, but it can still be seen that even in the age groups of 20 to 30 years old there are high figures of overweight people, ranging from 20% to 50% in select countries. In order to keep a healthy weight, to reduce the risk of diseases and to improve our mental health one of the most important things to do are physical exercises. However, only doing physical exercises is not enough. Tracking food consumption can give insights into the habits of different persons, whether good or bad. According to World Health Organization (WHO) a healthy diet is essential for good health and nutrition. It protects against many chronic noncommunicable diseases, such as heart disease, diabetes and cancer [7]. It is important to monitor the calorie intake and nutrients intakes, such as the carbohydrates, proteins, fats and also salt and sugars. However, this is not the only thing that is necessary for achieving a healthy lifestyle. Going to the gym, cycling, running, swimming and any kind of physical activity compliments a healthy diet, and beside the good effects on our body, it is also great for our mind. According to the Mental Health Foundation [8], engaging in physical activity releases chemicals in the brain that enhance mood, improve concentration, and promote better sleep.

Software is playing an increasingly important role in healthcare, taking over evermore-critical functions with the goal of increasing quality of care to patients/populations at reduced costs [3]. As of 2022 over 6 billion people own a smartphone [4], making mobile applications accessible to everyone. The time spent on these devices can be used in a way that facilitates a healthier lifestyle, by monitoring and tracking the physical activity and diet plan.

Cloud computing and health monitoring is increasingly being employed together as the healthcare equipment's are giving proper monitoring and collection of patient health record [5]. Internet of things (IoT) has been widely used to collect patient information, the healthcare industry changing in major ways during the last few years. This technology relieves the burden of the user, allowing passive input of data sustaining user's engagement. The massive amounts of data generated by IoT devices has led to the increased popularity of big data analysis, cloud computing services offering the necessary tools to analyse the data. This has led to the revolution of Industry 4.0 characterized by increasing automation and the employment of smart machines and smart factories [6].

This article aims to present the design and implementation of a mobile application for fitness and dieting. Such an application can promote a healthier lifestyle, help users achieve their goals, and collect data to contribute to research in the healthcare field. The scientific contribution within this article is not only the support in collecting relevant data for healthcare analysis, but also the review of existing scientific similar work and the use of cloud and IoT technologies in the proposed approach.

2. Related Work

2.1. Mobile Applications in Healthcare

In today's world developing cross-platform mobile applications is getting more and more popular. For an application to reach as many users as possible it must be compatible with both iOS and Android operating systems. Mobile devices have more sensors and more capabilities than desktop computers and for any computing device that contains sensitive information and accesses the Internet, security is a major concern for both enterprises and end-users [9]. The application needs to ask the user for permission to access the different sensors such as the camera and it is important to not offer too many or too little permission to the application as this can cause security problems [1]. Jain et al. [10] present the key attributes for a quality mobile application including adaptability, flexibility, security, convenience, ease of use, productivity and information persistence.

Some researchers believe that software systems can be a key enabler technology to address the increasing costs and challenges of access to quality healthcare [3, 11]. The main challenges identified relate to performance indicators of the sector, the risks of implementing changes to the healthcare system, the complexity of verification and validation processes, the data engineering required because of the unstructured form of information, etc. Knowledge translation (KT) can solve these problems, implying that the process consists of selecting and contextualizing basic research results, synthesizing and deploying solutions leveraging these results and evaluating their effectiveness to inform the next iteration of KT [3]. One of the central design features is customizability, a developed mobile application being capable of adapting to each end user needs. Another key element is interactivity, users having incentive to interact with the application when receiving rewards, being motivated by them.

Another important aspect is the analysis of the factors influencing the continuous use of a health application [12, 13]. The identified list includes the perceived usefulness of the application, how the application improves the health performance, and its effectiveness in getting healthier. The subjective norms have a significant effect on continuous use, such as the opinions of those around the user, expecting him to be healthier and the flow experience, application developers should focus on setting smaller challenges for the users to achieve. Main features requested include symptoms and data streams tracking like pain levels, fatigue, reports, storing documents from doctor appointments, integration with other health tracking applications and smart feedback that analyses patterns and suggests actions.

Recent studies focus on the use of gamification in health applications to increase the engagement of the users [14, 15]. Consequently, the applications can include features such as progress bars, badges, feedback when achieving goals, etc. However, in some cases, gamification techniques could affect user's understanding of the real benefits of a healthy lifestyle.

2.2. Cloud Computing in Healthcare

The use of cloud computing can improve the quality of mental health service platforms through automatic collection of the users' psychological data, and with monitoring, tutoring and warnings to facilitate preventive measures. Some experiments show that the accuracy of users' psychological test rising up to around 75% from 55% [16]. The use of cloud technologies such as Google Cloud Firebase can facilitate real-time data analysis and, in combination with machine learning, supports the prediction of heart disease [17].

Chung and Fong [18] talk about mobile phones as the personal health information system. With the rapid evolution of mobile devices and their rapid deployment in the healthcare environment they can play a big role in helping us achieve a healthier lifestyle. They offer location tracking, allowing for immediate rescue in case of emergency, medication tracking and reminders, Body Mass Index (BMI) calculators and symptom checkers. Using the cloud platform all this information can be seamlessly analysed.

Special attention is also paid to the effects of IoT, Cloud Computing and Big Data in the eHealth sector [19]. The fourth industrial revolution, Industry 4.0, has been enabled by the advancements in the areas mentioned above. This has also led to advancements in the healthcare industry, enabling Healthcare 4.0. Positive aspects of IoT devices apply to healthcare almost fully, allowing automatic collection of health metrics necessary for patient monitoring and provide different alerts based on the data collected. Cloud computing is a fundamental enabler of Healthcare 4.0 influencing a Healthcare-as-a-Service mindset, being useful for healthcare operators for data analysis, diagnosis and communication as well as for providing the services to the patients. Big data allows for extracting relevant information from amounts of data that were previously too big. They allow moving from research questions to predictive ones, aiming to find and solve potential problems much earlier than before.

3. Application Design

3.1. Technologies Used

In the development of the mobile health application the technologies opted for are the Flutter software development kit for the frontend and Firebase for the database and the backend services. Flutter allows for creating a multiplatform application from a single codebase, with good performances and an intuitive and efficient user interface created using widgets. OpenFoodFacts [20], an open-source database which contains detailed information about over 3 million food products has been used to allow users to find their consumed food products through its Application Programmable Interface (API).

Firebase is a Backend as a Service (BaaS) solution offered by Google, hosted in cloud offering support for many platforms. Its main advantages include always online status, an authentication system that is integrated with a real-time database, support for multimedia files storage and security rules for data protection [1]. In a mobile health application, it is important for the user to always have access to his data and to not lose any important information.

The authentication system offers support for this and together with Cloud Firestore, a NoSQL database Firebase offers all the necessary tools for developing the mobile application. The NoSQL database allows for storage of large amounts of records due to the collections and documents structure. In a health application data can take different structures, depending on the recorded parameters and a NoSQL database is perfect for this use-case. The Firebase services satisfy security and privacy standards, having completed ISO and SOC evaluation processes. Realtime Database Security Rules determine who has read and write access to the database [21], and through authentication we can authorize the users' access to data in the database. We can control his read or write access to different tables and paths and also apply validations to the data inserted in it.

3.2. System Design

The mobile application is built on the Model-View-ViewModel (MVVM) architecture [22]. It allows for the separating of logic between the application logic and the presentation logic. The Model contains the application data, the View defines what the user sees on the screen and handles users' inputs and the ViewModel is the component that connects the other two. It contains the properties to which the View is connected, and it handles the user actions, changing the data in the Model. The use case diagram (Fig. 1) presents the interactions between user and the system.

Fig. 1. The use case diagram of the proposed mobile application.

4. Relevant Features

This section presents the most important features of the application, how the user interacts with them and how they help him in achieving his goals and living a healthier lifestyle. The application is broken down into two modules, fitness and diet, and the user can choose to use only one of them, but the modules work together to bring him an even better experience.

4.1. Creating a Diet Plan

After creating an account and successfully logging in, in the diet module, the first action that the user needs to do is create a diet plan. Fig. 2 presents the form that the user needs to complete to generate a diet plan that suits his needs. Depending on his age, weight, height, activity levels, the desired diet type, gaining, losing or maintaining weight, there are multiple calculation methods to compute the calorie target per day for the user to reach his goal. Once the first diet plan has been created, the users' information is saved, and he can easily update it at any time. Fig. 3 shows the home screen with information about the users' day, including his calorie target, nutrient values. Depending on the type of diet the progress bar and the message are personalized to indicate how close the user is to reaching his goal for the day.

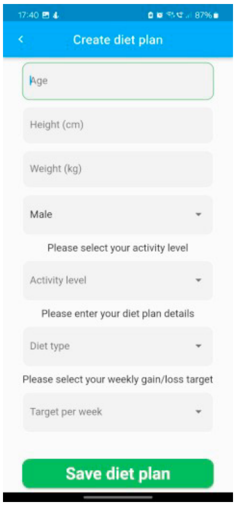

Fig. 2. Create diet plan form.

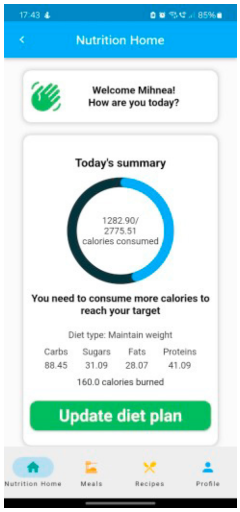

Fig. 3. Home page with relevant information for the user.

4.2. Managing your Meals

Once a user has created a diet plan, he can start managing his daily meals. The user can select the date for which he wants to see his summary and after that he can start adding meals. Information about calories and nutrients consumed are available (Fig. 4). Once a user adds a meal, he can go to the meal page by pressing on it. It contains the food products consumed by the user, with details about each one of them (Fig. 5).

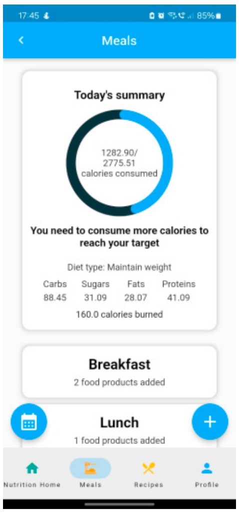

Fig. 4. Meals page.

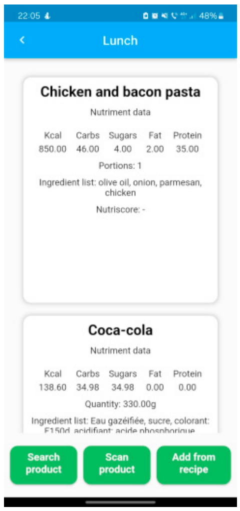

Fig. 5. Food products consumed in a meal.

From the meal page the user has three options to add a consumed food product, in order to cover all the possible cases. He can add consumed foods through the recipes added in the application, or using the data from the OpenFoodFacts database, by either searching the product by name (Fig. 6), or by scanning the barcode of the product. In all three cases he would need to add the quantity consumed, either servings, or in grams. Fig. 7, 8 and 9 [1] present the scanning of the barcode of a product, the identified product, and its nutritional values.

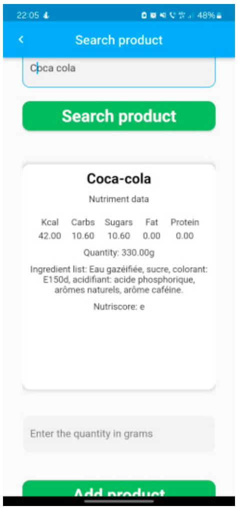

Fig. 6. Searching for a product.

Fig. 7. Scanning a product.

Fig. 8. Resulted product.

Fig. 9. Nutritional values of the product.

4.3. Recipes

The aim of this page is to make it easier for the user to track the consumed foods, especially if he follows a diet and has eating habits. A user can add his own recipes (Fig. 10) or can search for existing recipes (Fig. 11).

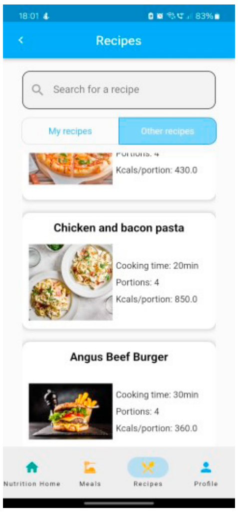

Fig. 10. The recipes page.

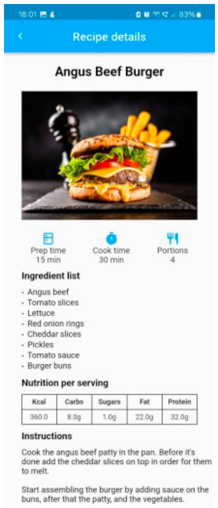

Fig. 11. Information about a specific recipe.

4.4. Workouts and Managing the Training Sessions

To start using the workouts module the user needs to select a training plan. He can either use one created by himself, or one created by other users and approved in the application. Once he does that he will see a weekly summary about his trainings, and how close he is to reaching his goals.

Once a user has set a workout plan, he can manage his training sessions. He needs to create trainings, and for each training session he can add the different exercises. In Fig. 12 the weekly summary is presented, with the training sessions for that week, and in Fig. 13 the exercises from one training session. Depending on the type of exercise the user has executed he needs to complete different fields. For a strength exercise he should enter the number of sets, reps and the weights used for the exercise while for a cardio exercise he needs to enter the distance covered. The differences are presented in Fig. 14 and 15.

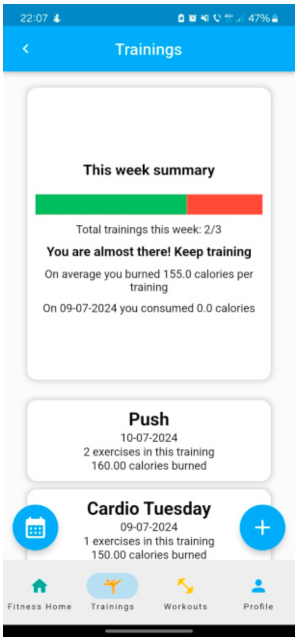

Fig. 12. Weekly training summary.

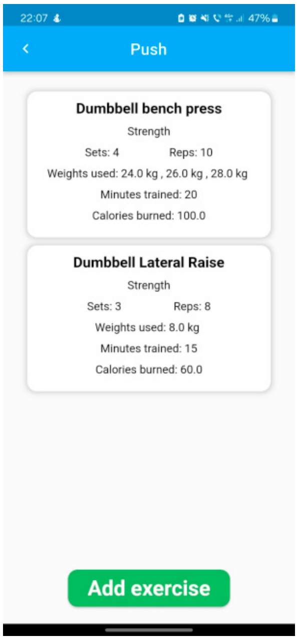

Fig. 13. Exercises from a training session.

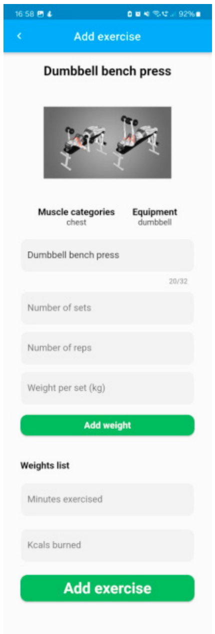

Fig. 14. Adding a strength exercise.

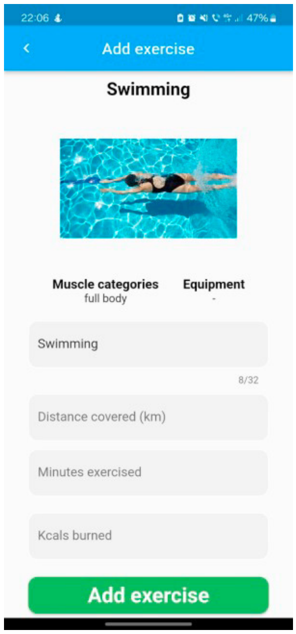

Fig. 15. Adding a cardio exercise.

5. Conclusions

With the new advancements in the IoT industry, mobile apps in the healthcare sector are becoming more and more useful, however, it's important to show this to the larger population. Nowadays, as everyone spends a big part of their day on the phone, it's important to give the people the tools they need in order to lead a healthy lifestyle. Mobile apps need to be built in such a way that they keep users connected and focused on achieving their goals. Many such apps can easily bore the persons using them, so it's important to make them as interactive as possible to keep users engaged.

The massive amounts of data generated by these kinds of applications can be used for medical research, population health analysis using cloud computing and machine learning. The impact of big data in healthcare is huge and as a future direction of research the ways in which it can be used must be investigated. This can lead to better patient care, improved research, reduce costs and personalized treatment plans.

Future research can focus on the optimization of data processing and analysis of user satisfaction according to existing methodologies such as Kano, or importance questionnaires to assess the customer perception related to the proposed features. The Kano methodology would help to identify the features of a product or a service that will delight potential customers [23].

References

[1] Dan, M.-V. (2023). *FitTrack – aplicație mobilă pentru monitorizarea activității fizice și a dietei* (Diploma project). Politehnica University of Timisoara, Timisoara.

[2] Sassi, F., et al. (2009), "The Obesity Epidemic: Analysis of Past and Projected Future Trends in Selected OECD Countries", OECD Health Working Papers, No. 45, OECD Publishing, Paris, https://doi.org/10.1787/225215402672.

[3] Weber-Jahnke, J. H., Price, M., and Williams, J. (2013). Software engineering in health care: Is it really different? And how to gain impact. 2013 *5th International Workshop on Software Engineering in Health Care (SEHC)*, 1–4. San Francisco, CA, USA: IEEE. https://doi.org/10.1109/sehc.2013.6602469.

[4] How Many People Own Smartphones? (2024-2029) | GilPress. (2024, January 31). Retrieved July 10, 2024, from https://whatsthebigdata.com/smartphone-stats/.

[5] Faridi, F., Sarwar, H., Ahtisham, M., Kumar, S., and Jamal, K. (2022). Cloud computing approaches in health care. *Materials Today: Proceedings*, **51**: 1217–1223. https://doi.org/10.1016/j.matpr.2021.07.210

[6] What is Industry 4.0 and how does it work? | IBM. (2021, August 4). Retrieved July 07, 2024, from https://www.ibm.com/topics/industry-4-0.

[7] Healthy diet. (n.d.). Retrieved July 10, 2024, from https://www.who.int/initiatives/behealthy/healthy-diet.

[8] Physical activity and mental health. (n.d.). Retrieved July 10, 2024, from https://www.mentalhealth.org.uk/explore-mental-health/a-z-topics/physical-activity-and-mental-health.

[9] Squires, W., and Centonze, P. (2016). Cross-platform access-rights analysis of mobile applications. *Proceedings of the International Conference on Mobile Software Engineering and Systems*. Presented at the ICSE '16: 38th International Conference on Software Engineering, Austin Texas. Austin Texas: ACM. https://doi.org/10.1145/2897073.2897717

[10] Jain, P., Sharma, A., and Aggarwal, P. K. (2020). Key attributes for a quality mobile application. *2020 10th International Conference on Cloud Computing, Data Science & Engineering (Confluence)*, 50–54. Noida, India: IEEE. https://doi.org/10.1109/confluence47617.2020.9058278.

[11] Birkhoff, S. D., and Moriarty, H. (2020). Challenges in mobile health app research: Strategies for interprofessional researchers. *Journal of Interprofessional Education & Practice*, **19**. https://doi.org/10.1016/j.xjep.2020.100325.

[12] Yan, M., Filieri, R., Raguseo, E., and Gorton, M. (2021). Mobile apps for healthy living: Factors influencing continuance intention for health apps. *Technological Forecasting and Social Change*, **166**. https://doi.org/10.1016/j.techfore.2021.120644.

[13] Polhemus, A., et al. (2023). Experiences of health tracking in mobile apps for multiple sclerosis: A qualitative content analysis of user reviews. *Multiple Sclerosis and Related Disorders*, **69**. https://doi.org/10.1016/j.msard.2022.104435

[14] Ansong, S., and Iahad, N. A. (2019). A conceptual model for designing a gamified intermittent fasting mobile app. *2019 6th International Conference on Research and Innovation in Information Systems (ICRIIS)*. Presented at the 2019 6th International Conference on Research and Innovation in Information Systems (ICRIIS), Johor Bahru, Malaysia. Johor Bahru, Malaysia: IEEE. https://doi.org/10.1109/icriis48246.2019.9073629.

[15] Pernencar, C., Sousa, P., Frontini, R., Martinho, R., Runte, D., Mendes, D., and Carvalho, M. (2018). Planning a health promotion program: Mobile app gamification as a tool to engage adolescents. *Procedia Computer Science*, **138**: 113–118. https://doi.org/10.1016/j.procs.2018.10.016.

[16] Han, X., Bai, L., and Han, B. (2023). Evaluation and application of cloud computing in mental health service platform. *Procedia Computer Science*, **228**: 1160–1167. https://doi.org/10.1016/j.procs.2023.11.151

[17] Desai, F., et al. (2022). HealthCloud: A system for monitoring health status of heart patients using machine learning and cloud computing. *Internet of Things*, **17**. https://doi.org/10.1016/j.iot.2021.100485.

[18] Chung, W. -Y., and Fong, E. M. (2014). Seamless personal health information system in cloud computing. *2014 36th Annual International Conference of the IEEE Engineering in Medicine and Biology Society*, **119**: 3658–3661. Chicago, IL: IEEE. https://doi.org/10.1109/embc.2014.6944416.

[19] Aceto, G., Persico, V., and Pescapé, A. (2020). Industry 4. 0 and health: Internet of things, big data, and cloud computing for healthcare 4. 0. *Journal of Industrial Information Integration*, **18**. https://doi.org/10.1016/j.jii.2020.100129.

[20] Openfoodfacts. Org. (n.d.). Retrieved July 10, 2024, from https://world.openfoodfacts.org.

[21] Firebase | Google's mobile and web app development platform. (n.d.). Retrieved July 10, 2024, from Firebase website: https://firebase.google.com/.

[22] Stonis, M. (2022). Enterprise Application Patterns using .NET MAUI. Microsoft Developer Division.

[23] Pop, M.-D., Pugna, A. P., Crețu, V.-I., and Potra, S. A. (2022). Two years of hybrid education in Romania: A comparative overview of the students' expectations for the online educational platforms. *Sustainability*, **14 (21)**. https://doi.org/10.3390/su142114241.